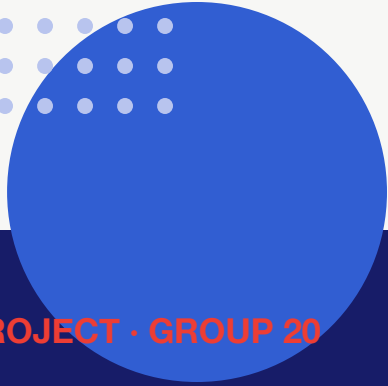




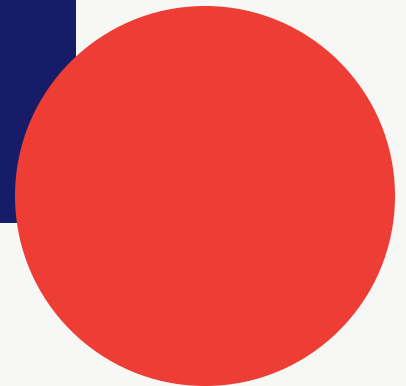
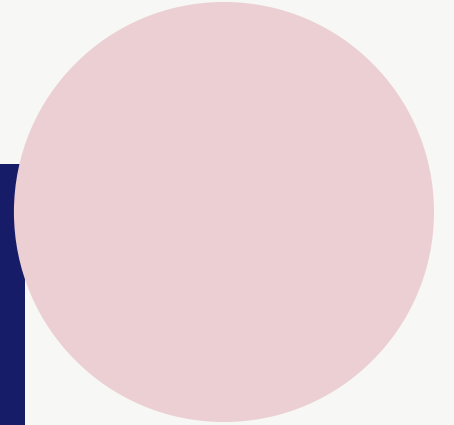
MSC GROUP PROJECT · GROUP 20

London Sport 2

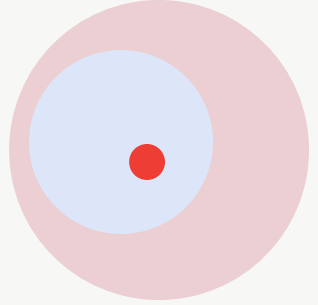
Forecasting physical activity participation, volunteering
and activity patterns in London

Siyan Xin · Jingyi Hua · Shuhan Zhao · Yifeng Mao

Supervisor: Dr Dalila O'Grady | 4 September 2026



Presentation route



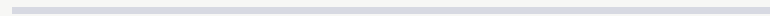
1

Introduction



2

Background



3

Common methods



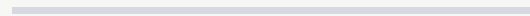
4

Four research questions

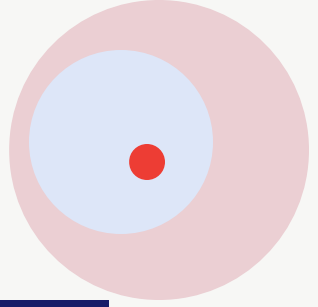


5

Summary



6



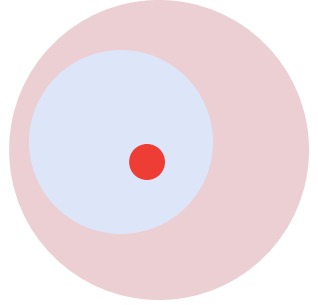
1

Introduction



London-wide averages can hide local need

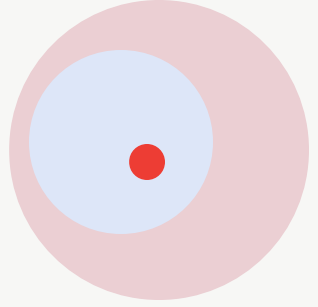
Historical monitoring tells us where participation is now. Planning requires evidence about where gaps may persist, narrow or widen.



THE DECISION PROBLEM

Plan across place, people, settings and volunteering – without treating forecasts as certainties.

One survey supports four connected planning questions



RQ1

Place

London, Inner/Outer and borough trends

RQ2

People

Age, disability and specific activities

RQ3

Settings

Indoor/outdoor balance and activity level

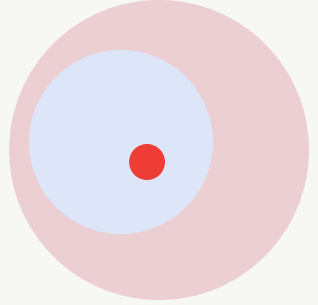
RQ4

Community

Frequent sport volunteering



Broad coverage, bounded claims



2015/16

2022/23

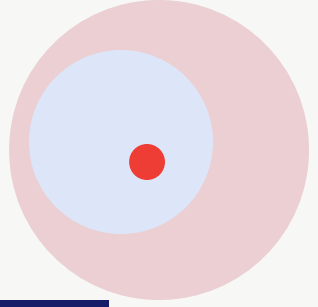


RQ1–RQ2 · eight waves · eight-year forecasts

RQ3–RQ4 · six waves · six-year forecasts

Adults 16+ · London overall · Inner/Outer London · 32 boroughs

Repeated cross-sections + short time series = conditional planning scenarios, not causal estimates.

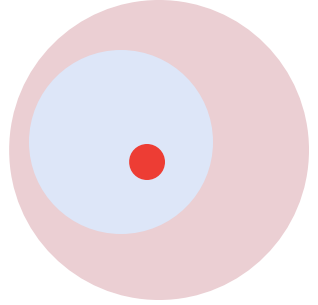


2

Background

What previous evidence tells us—and where local forecasting remains limited.





Previous research identifies unequal participation patterns

Representative evidence is grouped, with full details kept in the dissertation bibliography.

SOCIOECONOMIC AND LOCAL CONDITIONS

Physical inactivity varies by socioeconomic position, while London evidence links activity to neighbourhood conditions in deprived communities.

AGE AND LIFE COURSE

Activity profiles differ across age groups, and older-adult participation is shaped by age, deprivation and perceived opportunity.

DISABILITY AND ACCESS

Disabled adults report different activity types and durations; barriers include facilities, transport, support and the organisation of sport.

ACTIVITY ENVIRONMENTS

Green space, built environment and active-travel infrastructure shape the opportunity to be active.

VOLUNTEERING

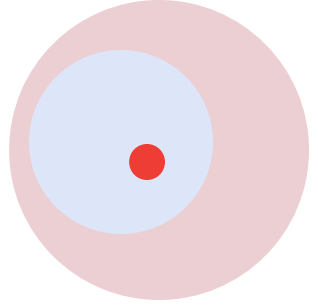
Sport participation is associated with sport volunteering and supports local clubs and programmes.

COVID DISRUPTION

Pandemic restrictions disturbed activity opportunities, so recent waves need cautious interpretation.

KEY MESSAGE

Prior research establishes the main participation inequalities, but it does not provide local multi-year forecasts for London.



Existing work defines the forecasting safeguards

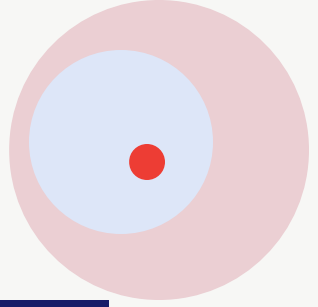
The methodological background explains why the project uses baselines, valid transformations and later-year evaluation.



DESIGN IMPLICATION

Use chronological validation, task-specific metrics, valid probability transformations and explicit sensitivity checks.

KEY TIP: The methods are chosen to reduce leakage, invalid shares and overconfident long-horizon interpretation.



3

Common methods

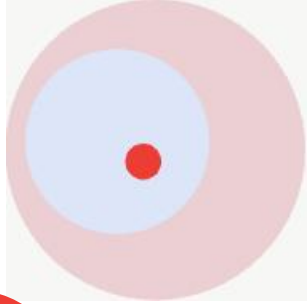
One survey pipeline, task-specific outcomes, chronological evaluation.





From survey responses to analysis data

All four research questions use the same preparation rules



01

Load survey data

Eight annual survey waves

02

Keep London

Adults in 32 boroughs

03

Match variables

Same names and codes
each year

04

Apply weights

Results represent London
adults

05

Create datasets

One longer, one more
detailed

LONGER DATASET

Used for RQ1 and RQ2
2015/16 to 2022/23
135,497 people 124 comparable
activities

DETAILED DATASET

Used for RQ3 and RQ4
2017/18 to 2022/23
96,629 people 179 available activities

Coding rules

A real zero means the person did not take part. Blank, skipped or unavailable answers stay missing.

Monthly or quarterly timing is available for 134,916 records. City of London is not included in borough models because its sample differs.

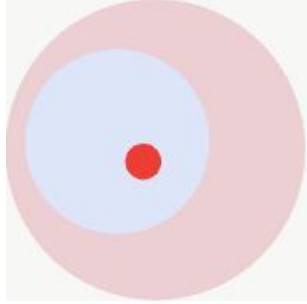
WHY THIS MATTERS

The preparation rules stop missing answers from being mistaken for no activity.



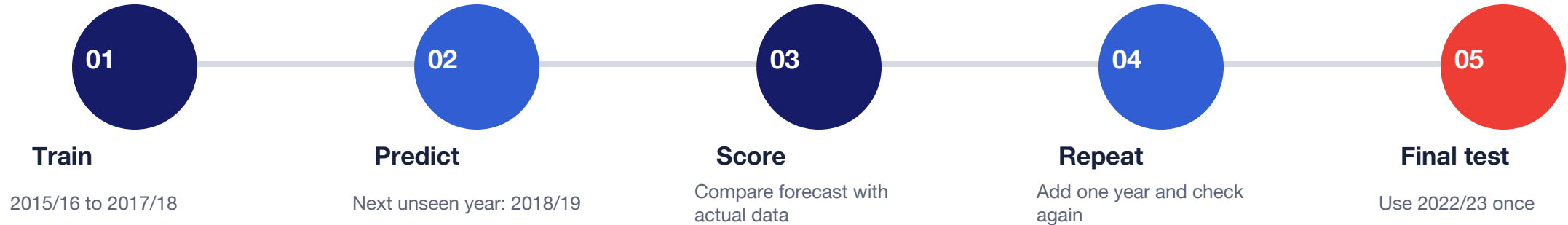


How the forecast model was chosen and tested



The model always learns from earlier years and predicts a later year

ROLLING VALIDATION

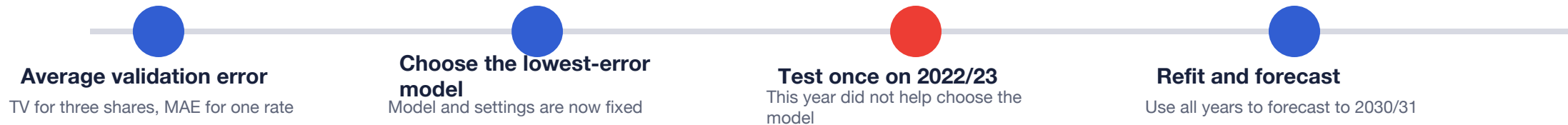


RQ1 and RQ2 use four validation years ending in 2021/22. RQ3 and RQ4 use three because their data start in 2017/18.

FOUR MODELS COMPARED

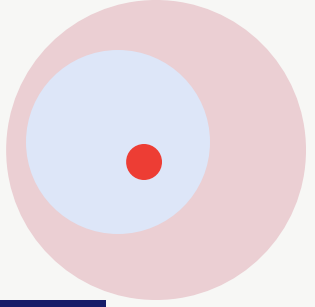


HOW THE WINNER IS USED



WHY THIS MATTERS

The model never sees future data during selection, so the final test reflects real forecasting conditions.



4

RQ1 · Place

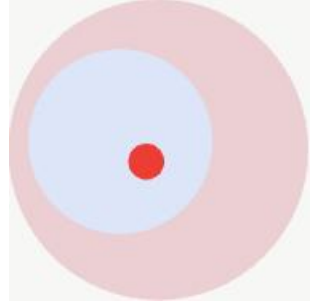
Physical activity trends, borough differences and forecasting.



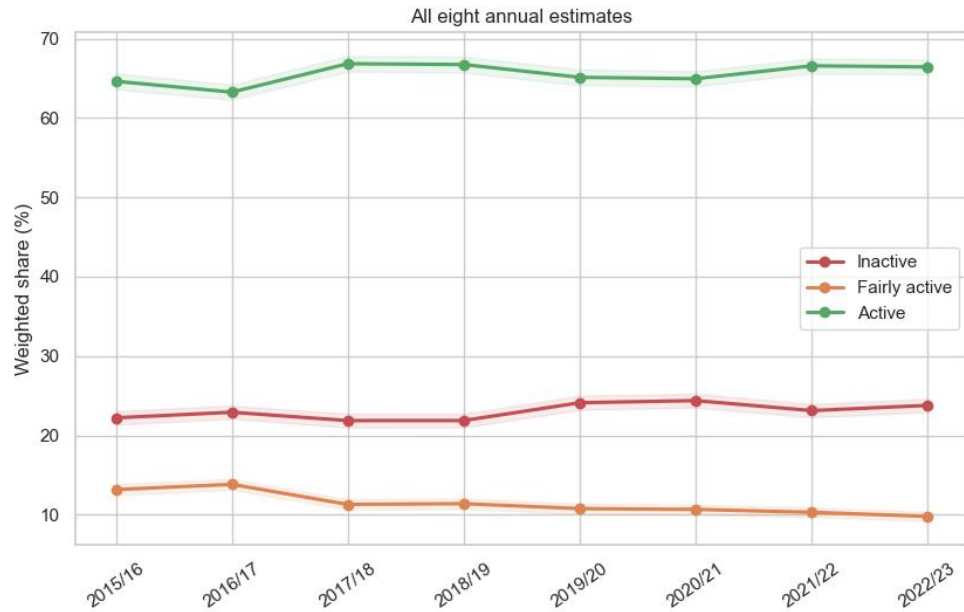


London activity changed in different directions

Methods: survey weights, seasonal decomposition, age adjustment and weighted regression



London-wide activity states



London-wide Active

+1.8 pp

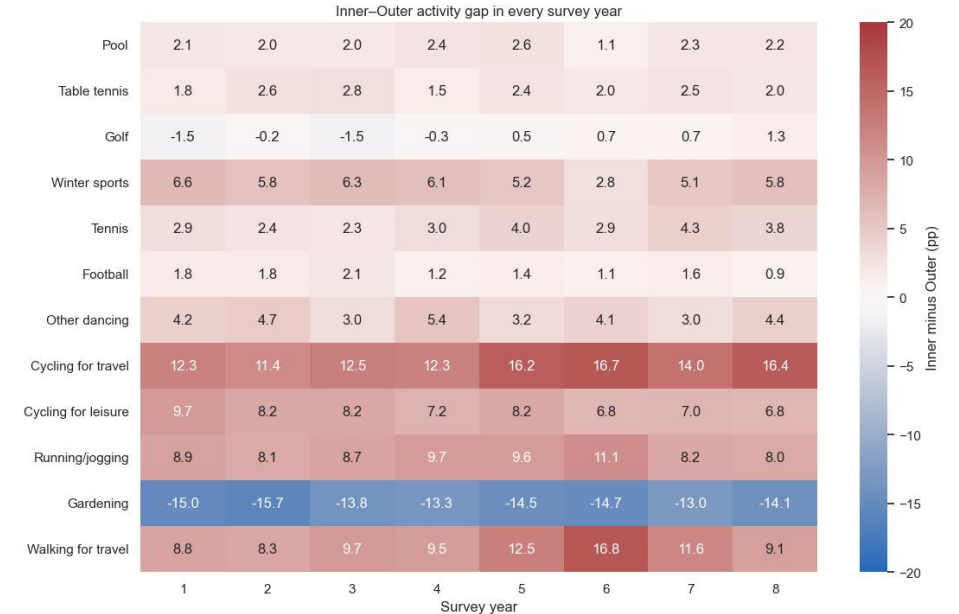
London-wide Inactive

+1.6 pp

London-wide Fairly Active

-3.4 pp

Inner minus Outer London by activity



Inner London Active vs Outer

+5.6 pp raw +4.5 pp adjusted

STL attributes 55.4% of monthly variation to seasonality and 8.0% to the long-term trend. Pandemic adjustments are associational.

KEY MESSAGE

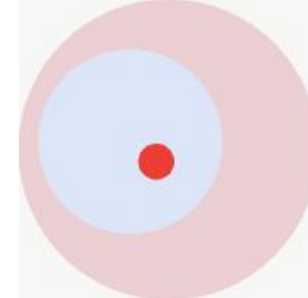
London moved towards both Active and Inactive, while Inner and Outer London kept different activity profiles.



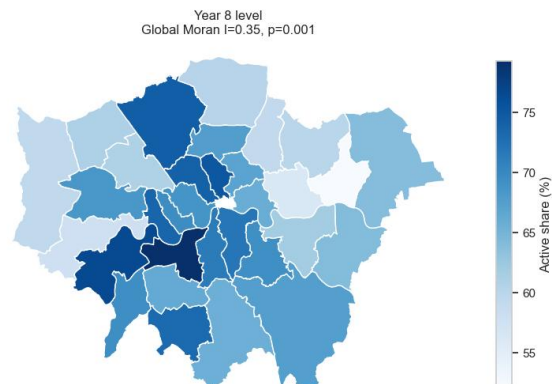


Borough differences form stable spatial groups

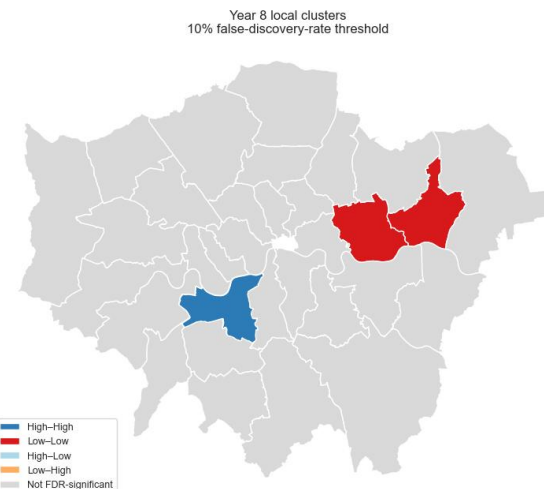
Methods: rank correlation, mixed modelling, Local Moran's I, PCA, K-means and 500 resamples



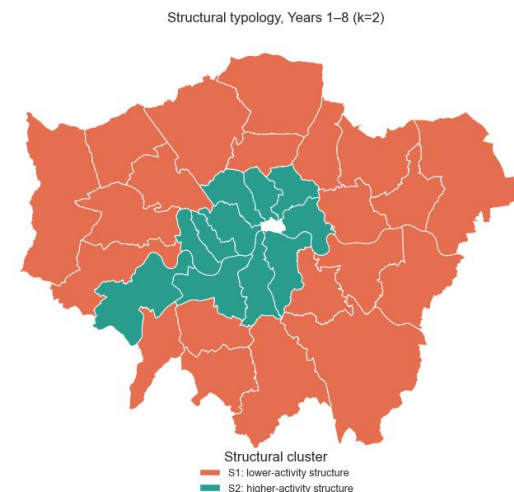
Activity level in 2022/23



Significant spatial clusters



Two structural groups



Persistence

Rank correlation 0.71

Between-borough variation

75.8%

Spatial clustering

Moran's I 0.35 p = 0.001

Stable grouping

k = 2 ARI 1.000

Wandsworth is High-High. Newham and Barking and Dagenham are Low-Low after false discovery rate adjustment. The structural solution contains 21 lower and 11 higher boroughs.

KEY MESSAGE

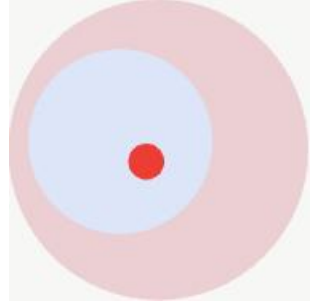
Borough differences remained large, spatially clustered and structurally stable.



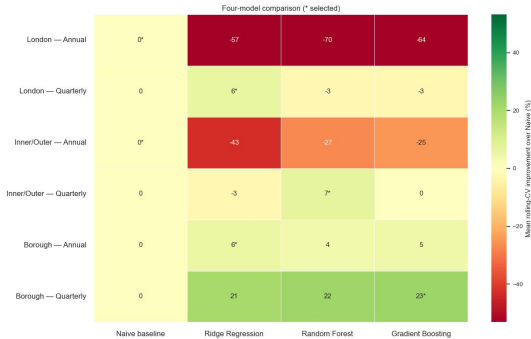


Forecasts guide annual planning

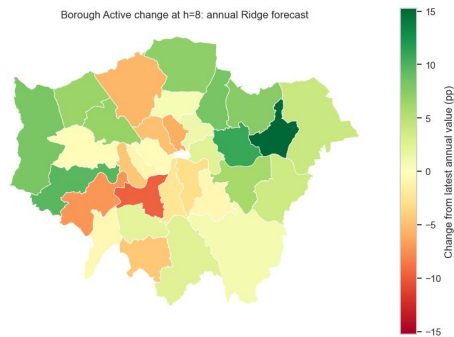
We compared four models across annual, quarterly and monthly tasks, then tested the winner on 2022/23



Model comparison

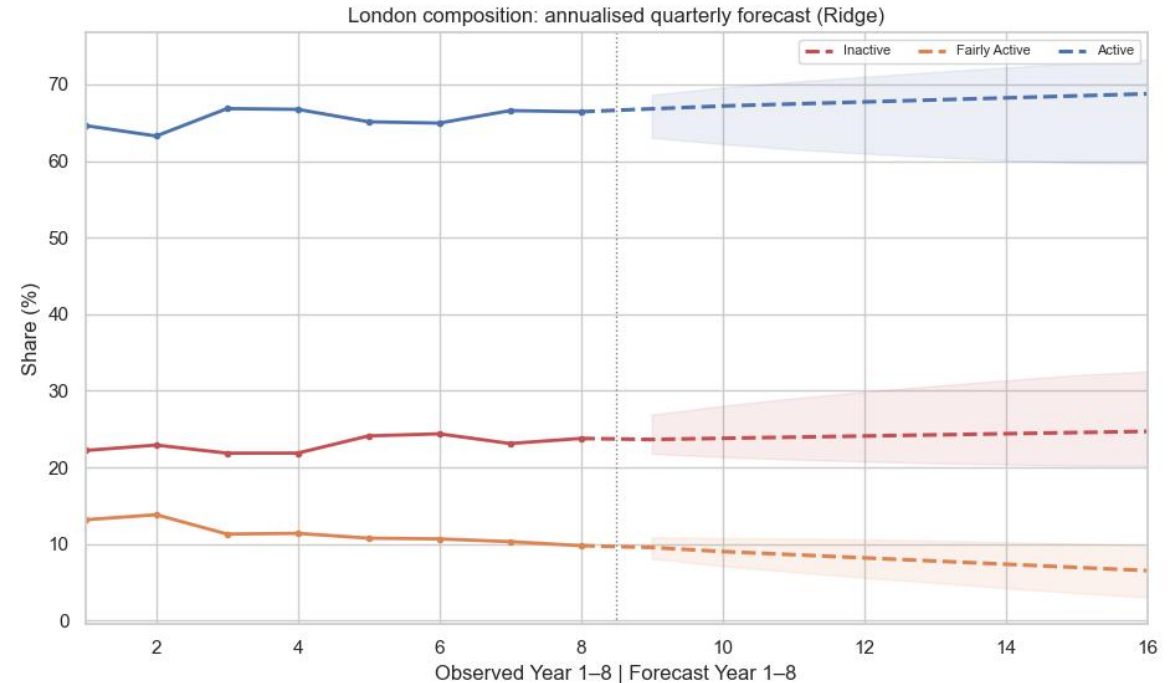


Eight-year borough change



Median +1.1 pp Range -9.7 to +15.3 pp

Annualised quarterly London scenario



Annual models: Naive for London and Inner/Outer, Ridge for boroughs.

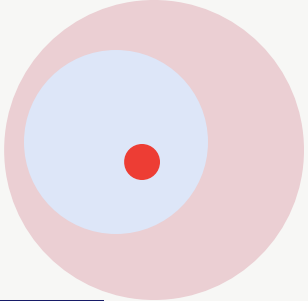
Quarterly London scenario by 2030/31: Active +2.0 pp, Fairly Active -3.2 pp, Inactive +1.2 pp.

The shaded bands show sensitivity, not formal prediction intervals.

KEY MESSAGE

Annual forecasts provide the main evidence. The quarterly result is a cautious scenario, not a firm prediction.



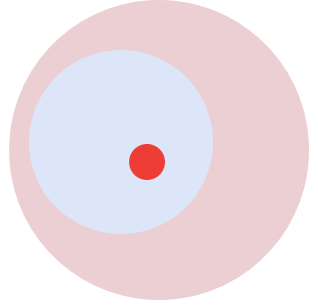


4

RQ2 · People

Forecasts by age, disability and specific activity.





Eight tasks require RQ2-specific reliability checks

The common validation framework is not repeated here; the focus is on decisions specific to age, disability and activity forecasting.

8

age groups

16–24 to 85+

3 +
13disability
measuresbroad groups +
overlapping indicators

124

activities

comparable across
eight waves

8

forecasting tasks

4 outcomes × 2
strands

OUTCOMES PER STRAND

Overall composition

Inactive / Fairly Active / Active

Activity composition

same three shares by activity

Past-12-month

participation reach

Regular participation

sustained engagement

RQ2-SPECIFIC METHOD ARCHITECTURE

Compositional modelling

ALR with Active reference; inverse
transform; ϵ sensitivity

Activity ranking

predicted participation rates; near-ties
flagged

Recursive evaluation

2023/24–2030/31 forecasts; H1–H4
historical backtests

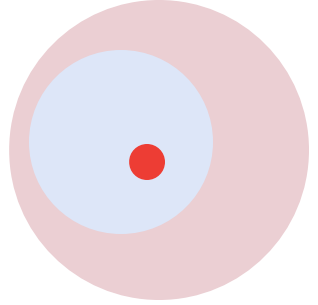
Robustness layer

neff diagnostics; shrinkage; residual
simulation; model, weight, ϵ and COVID
sensitivity

KEY MESSAGE

The strongest RQ2 contribution is not only forecasting groups, but making sparse, compositional and long-horizon results auditable.





Disability composition tasks showed the strongest test gains

Evaluation is presented before interpretation: each selected model must improve unseen-year prediction, not just fit history.

MODEL SELECTION

Gradient Boosting
selected for 6 / 8 tasks

Naive: age past-12-month

Ridge: age regular participation

UNTOUCHED 2022/23 TEST

29.1%

lower TV

Disability activity
composition

26.2%

lower TV

Disability overall
composition

21.4%

lower TV

Age overall
composition

RECURSIVE BACKTESTING

H1

H2

H3

H4



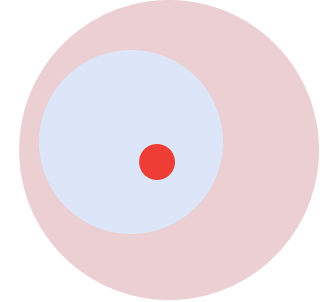
Errors were checked across directly testable H1–H4 horizons; longer horizons are treated more cautiously.

WORDING TO DEFEND

Complexity was retained only when validation supported it; historical backtests directly support H1–H4, not all eight future years.

KEY MESSAGE

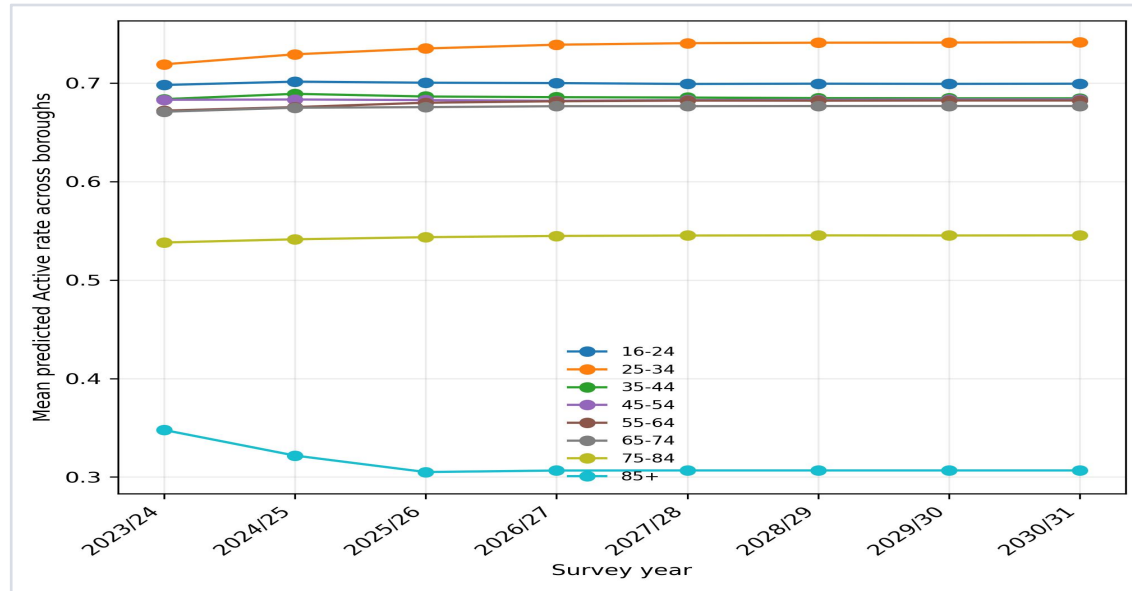
The test evidence is strongest for disability compositions; long-horizon precision is deliberately not over-claimed.



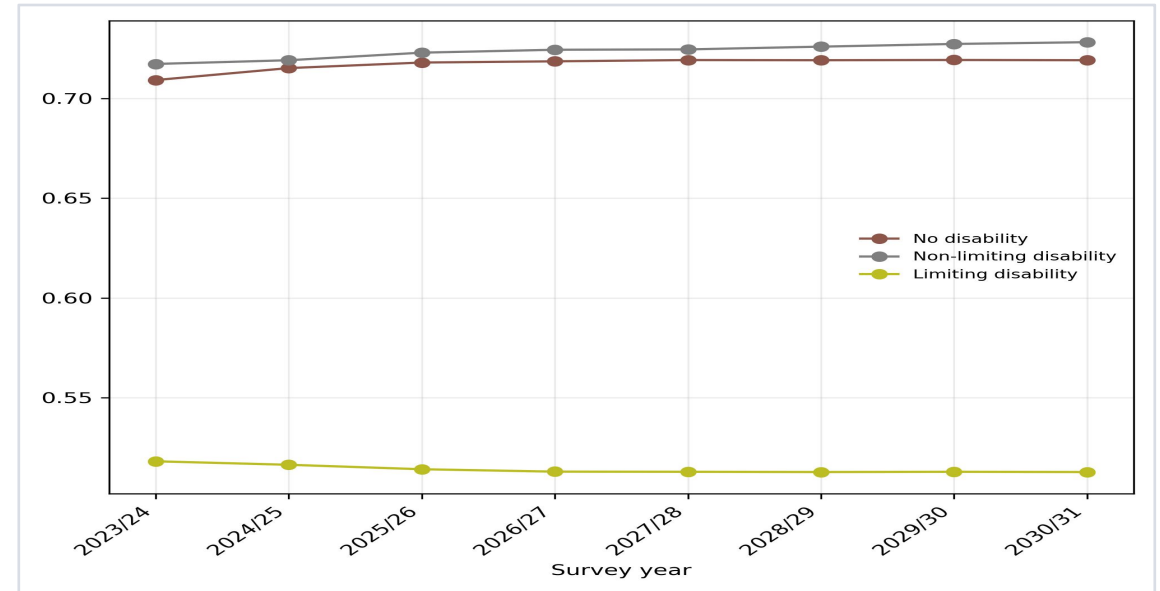
Older and limiting-disability groups remain at lower predicted Active rates

The forecast interpretation separates robust relative differences from model-sensitive long-run slopes.

AGE FORECAST



DISABILITY FORECAST



Age

85+ remains among the lowest predicted Active-rate groups.

Disability

Limiting disability remains below no disability.

Activities

Active Travel and Gardening often rank highly.

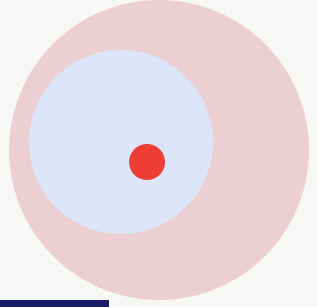
Robustness

Exact long-horizon slopes are more model-sensitive than the relative demographic pattern.

CAUTIOUS INTERPRETATION

Activity rankings reflect predicted participation rates, not stated preferences; near-ties should not be over-interpreted.

KEY MESSAGE Relative demographic differences are more robust than exact long-run trajectories.



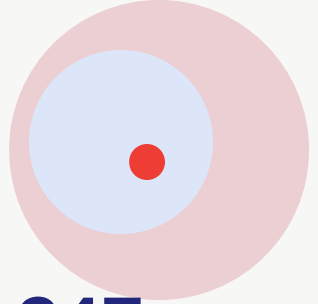
4

RQ3 · Settings

Indoor/outdoor participation forecasts and activity-level differences.



RQ3 creates a valid four-part outcome



6

waves

146

comparable activities

96,629

respondent-waves

576 / 6,847

annual / monthly cells

Qualifying participation

Months = 1 AND Days ∈ {1,2}

Setting is evaluated only after this rule is met.

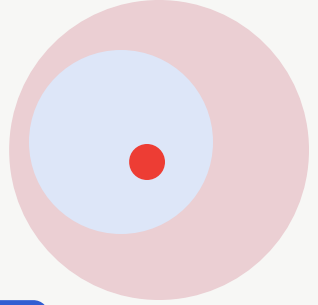
Indoor only

Outdoor only

Both

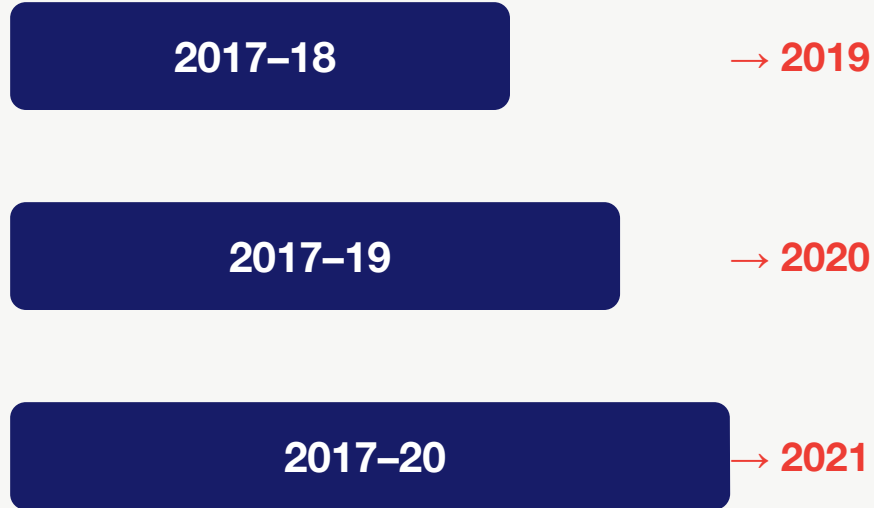
Neither recorded

‘Neither recorded’ is audited separately and does not imply physical inactivity.

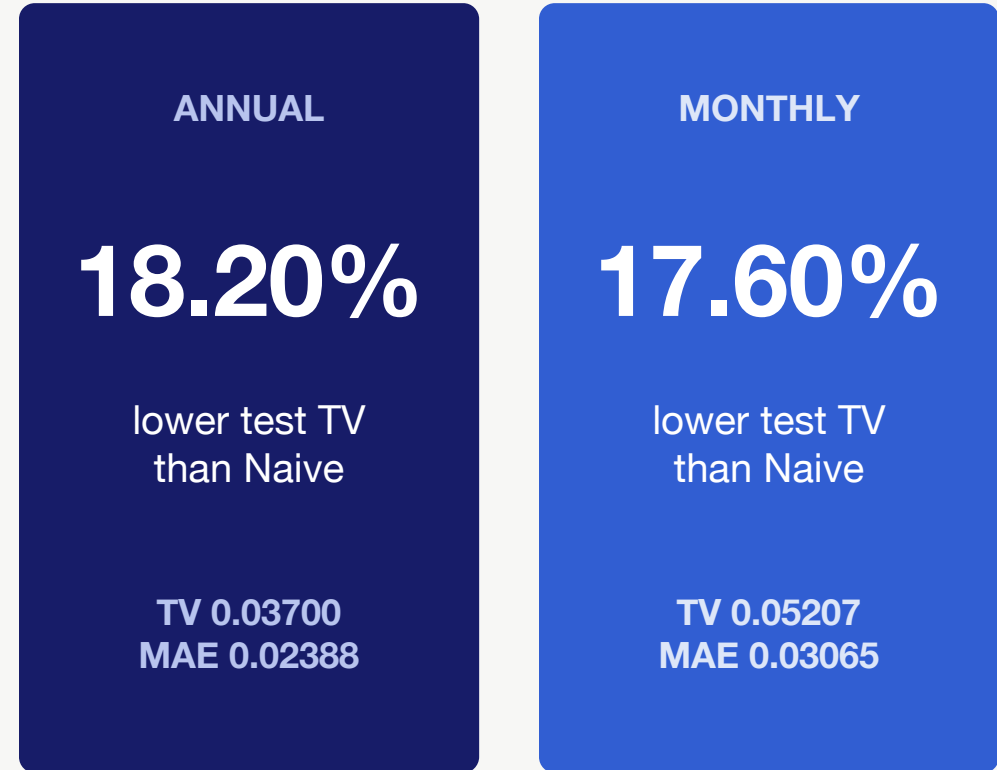


Chronological testing favoured Random Forest

Three expanding origins

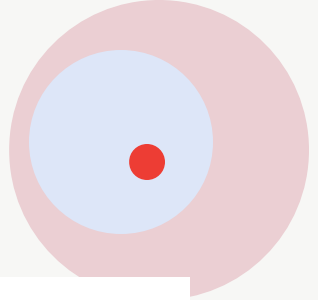


Untouched final test: 2022/23



Annual evidence is primary · 6,139 of 6,847 monthly cells have neff < 20

Indoor exposure stays above 50% in both scenarios



Observed

62.60%

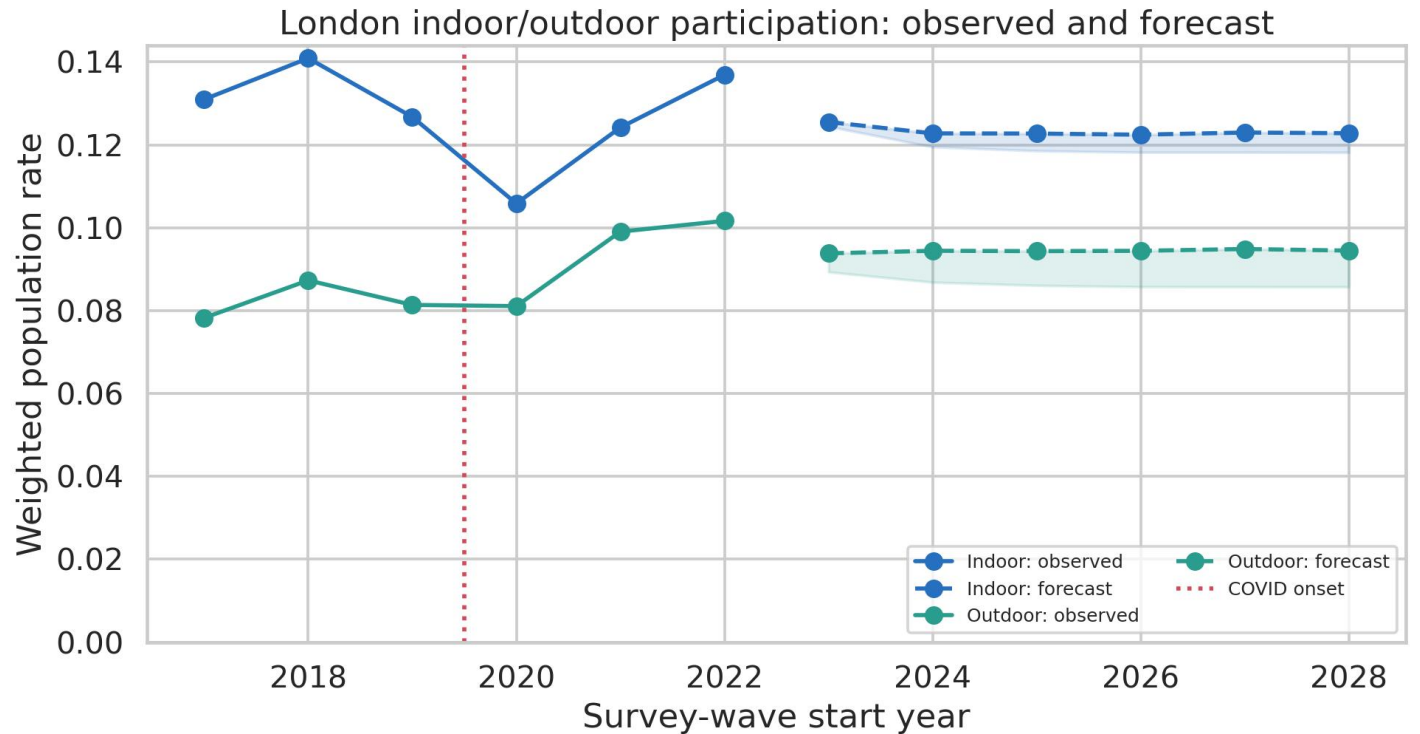
2017/18 indoor exposure



57.39%

2022/23 indoor exposure

Recorded-exposure balance, not a share of people.



Persistent legacy
56.52% in 2028/29

Legacy recovery
57.96% in 2028/29

Scenario range ≠ prediction interval

Less-active groups show a stronger indoor balance

Active – Inactive



-11.25
pp

95% CI -19.49 to -2.44

Active – Fairly Active



-9.12 pp

95% CI -12.33 to -6.06

Insufficiently Active – Inactive



-2.13 pp

95% CI -11.18 to 6.63

WHAT IT MEANS

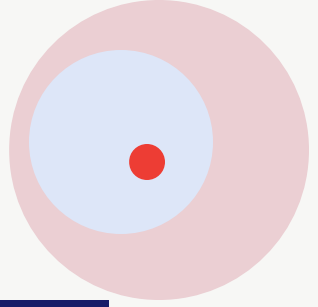
Active respondents show a lower indoor exposure balance than the other groups.

Not a three-level gradient; not a causal effect.

Among respondents with recorded indoor/outdoor exposure

Implication for London Sport

Maintain accessible indoor provision, while building gradual and locally validated pathways into outdoor activity.

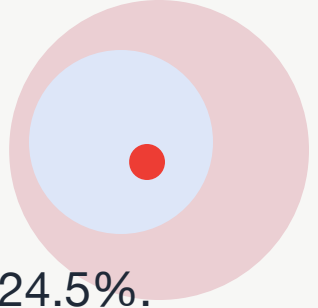


4

RQ4 · Sport Volunteering

Frequent sport volunteering rate across boroughs, ages and activity levels.





Three forecasting tasks

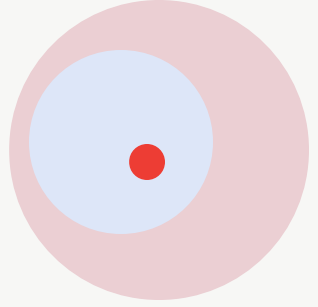
All three selected models outperformed the Naive baseline, reducing test error by 17.0%–24.5%.

Task	Selected model	2022/23 test MAE	Improvement vs Naive
Borough × Activity level	Random Forest	3.18pp	17.0%
Borough × Age	Gradient Boosting	5.01pp	24.5%
Borough × Age × Activity level	Gradient Boosting	6.83pp	23.5%

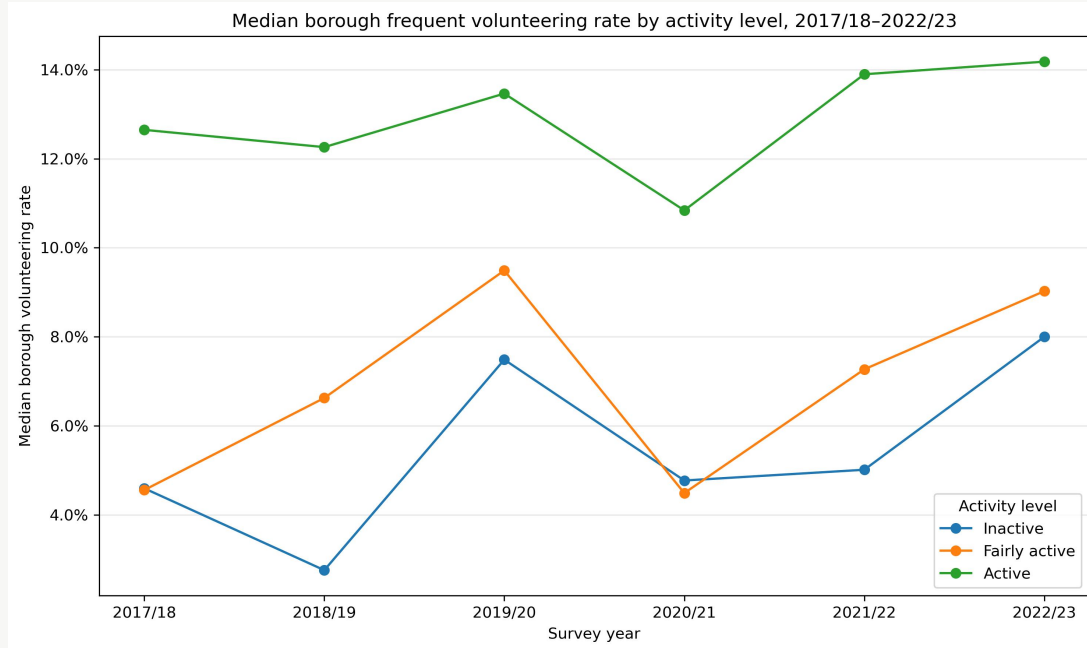
MAE = effective-n-weighted mean absolute error;

Model selection was based on rolling validation; 2022/23 was used only for independent testing.

Borough × Activity level

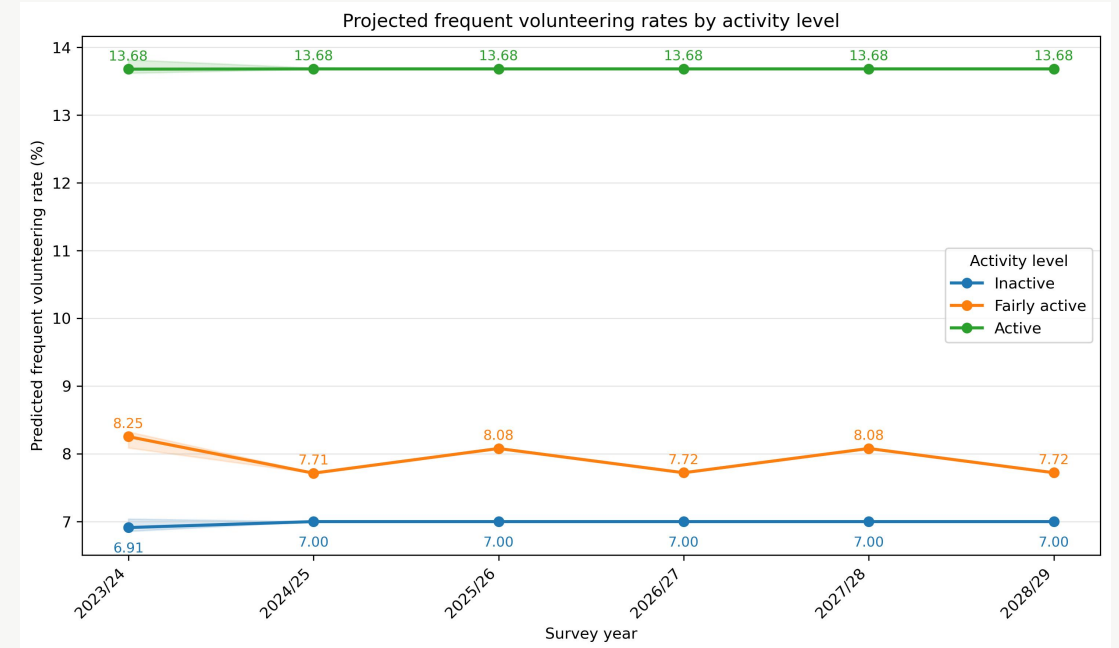


The historical activity-level ordering is preserved in the six-year forecasts.



Historical benchmark

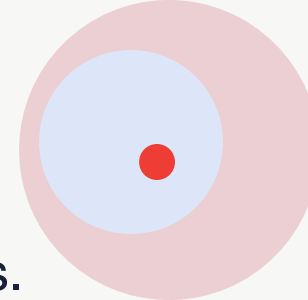
Active: 13.1%
 Fairly Active: 7.0%
 Inactive: 4.9%



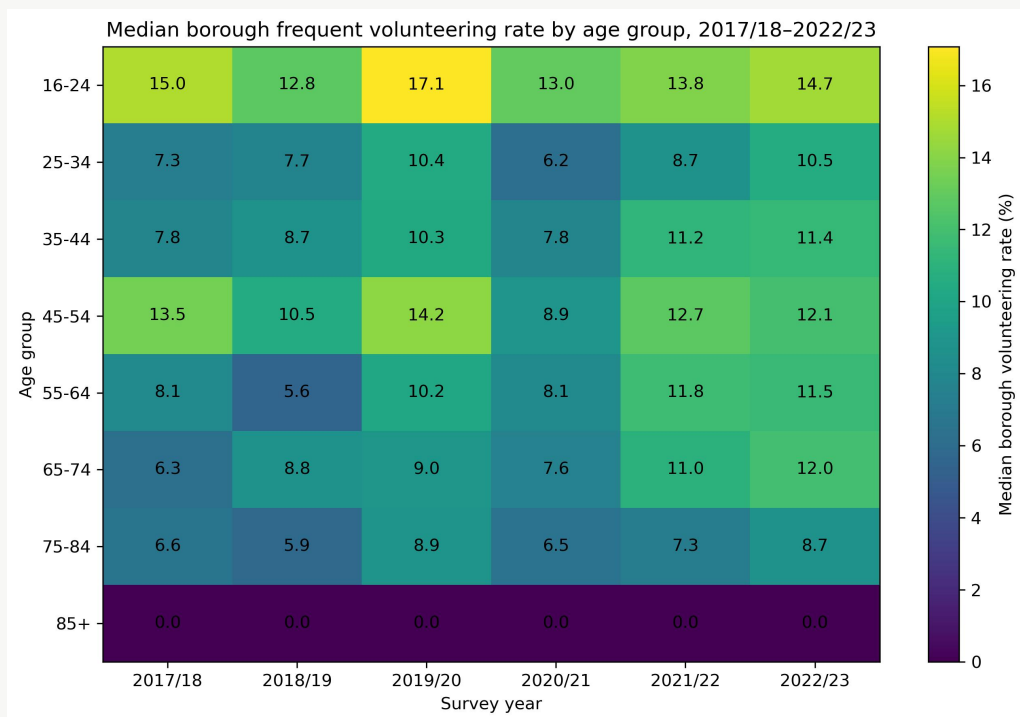
Forecast

Active ≈ 13.7%
 Fairly Active ≈ 7.7–8.3%
 Inactive ≈ 6.9–7.0%

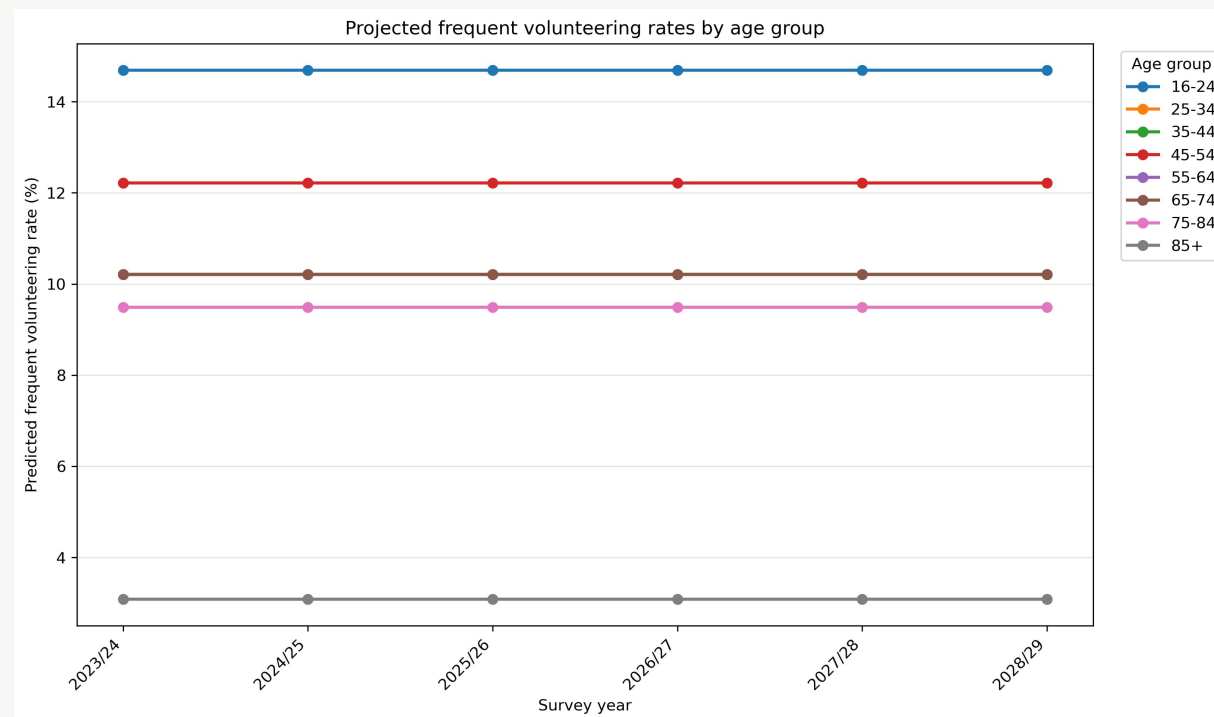
Borough × Age



Age differences persist in the forecasts, despite relatively flat projected trajectories.



Highest six-year historical median:
16–24 14.26%



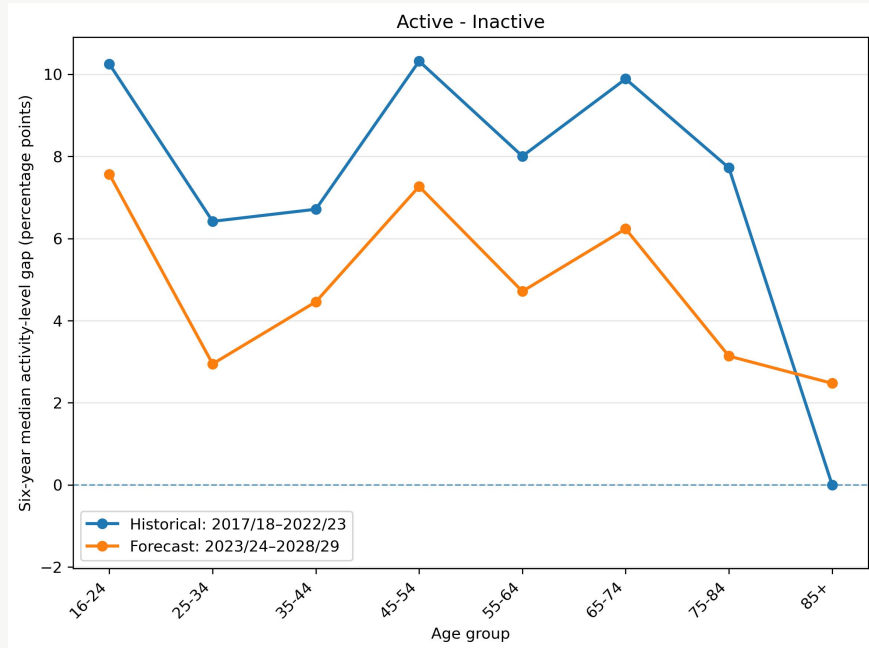
Highest: 16–24 14.69%

Lowest: 85+ 3.09%

Flat trajectories reflect the fitted model, not certainty of no future change.

■ The projected gaps between activity levels

The projected activity-level gap varies by age, mainly through the Active–Inactive contrast.

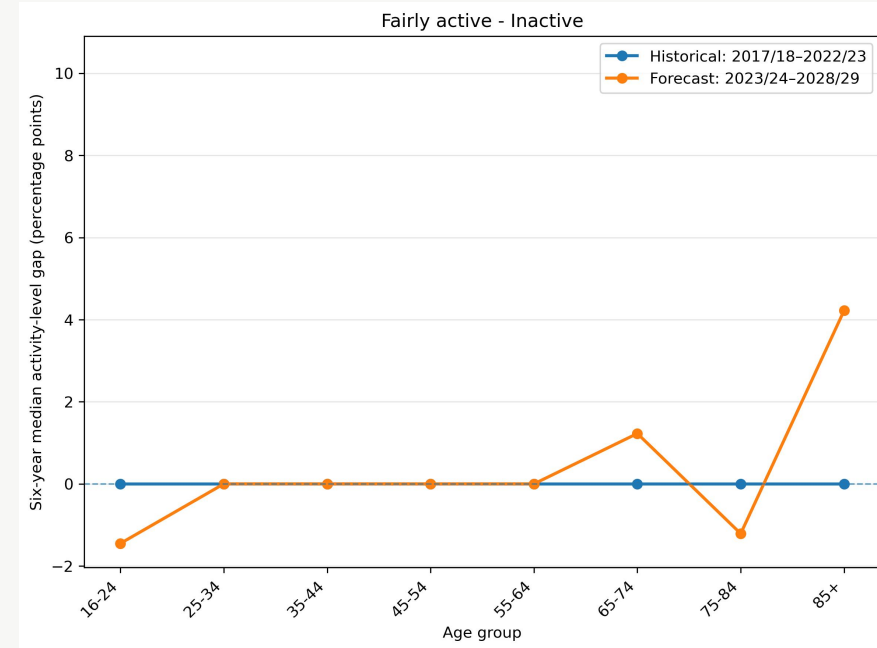


16–24 7.57 pp

45–54 7.27 pp

25–34 2.95 pp

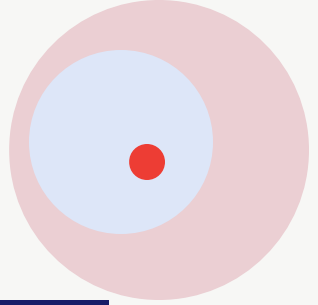
Generally smaller than historical gaps for ages 16–84



Close to zero for most ages

85+ ≈4.22 pp

- • • • • Gaps are calculated within the same borough and age group first, then summarised across boroughs and across the six historical or forecast years.



5

Summary

Main conclusions, limitation and future work.



Key findings across the project

01

Geographical Differences

Persistent borough inequalities;
No clear common long-term annual trend

02

Age and Disability

Older and limiting-disability groups remain disadvantaged

03

Indoor and Outdoor Activity

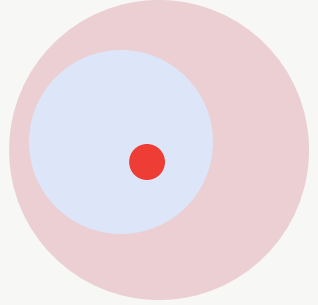
Indoor exposure remains dominant;
Active adults are less indoor-oriented

04

Sport Volunteering

Active adults show higher volunteering rates;
Active–Inactive gaps vary by age

Planning should therefore be targeted by place and population group, rather than based on a single London-wide trend.



Limitations and future work

Key limitations

Short historical series

6–8 annual survey waves

Sparse subgroup data

Sample support weakens after disaggregation

Limited long-horizon validation

Future work

Prospective validation

Test issued forecasts on new waves before retraining

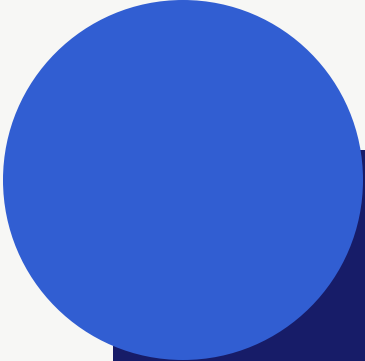
Stronger modelling

Multilevel / hierarchical / Bayesian approaches

Richer forecasting inputs

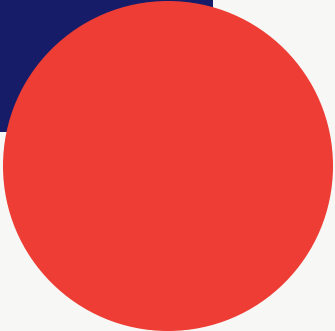
Reliable external predictors + population projections

Forecasts are conditional planning evidence, not precise or deterministic predictions.

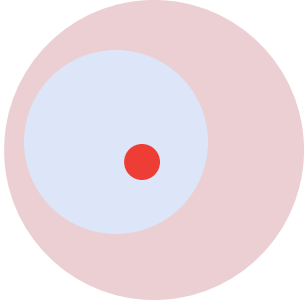


THANK YOU

London Sport 2 · Group 20



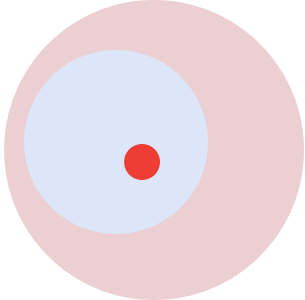
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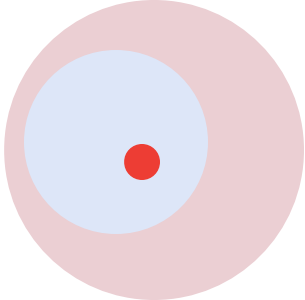
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